

SUPPLEMENTARY DOCUMENTATION

PhysMon Research Proposal

Change Log, Unresolved Decisions, Assumptions,
Compilation Notes, and Self-Audit

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1 Change Log: Major Scientific Changes from the Original Proposal

The following table documents the most consequential scientific changes made during the revision from “PhysTrace” to “PhysMon.” Changes are ordered by scientific importance.

Table 1: Major scientific changes.

Original design	Revised design (PhysMon)	Reason for change
[C1] Primary target: prompt condition. Probe was trained to classify “shortcut-available vs. shortcut-absent” prompt condition.	Primary target is now model-specific counterfactual shortcut sensitivity $S_{\theta}(\tau)$, behaviourally derived from output instability across solver-verified invariant families.	Prompt-condition probing is equivalent to detecting inserted cue text; it does not establish that the model relied on the cue. The Sahoo et al. format-confound critique applies directly to the original design.
[C2] No three-level distinction. “Shortcut-available” was used to implicitly mean both cue presence and shortcut-reliant computation.	Explicit three-level framework: (1) prompt condition, (2) observed model behaviour, (3) internal representation. These are never conflated.	These levels require different evidence and different claims. Conflating them is the most common and damaging reviewer objection for this class of paper.
[C3] Overclaimed causal framing. Title and claims stated “causal activation monitoring for shortcut reliance,” implying a recovered reasoning path.	Causal claim limited to: <i>localised representations contribute to cue sensitivity</i> . No claim about recovering the reasoning path or proving shortcut use in the full computational sense.	Recent reasoning-interpretability papers show that strong diagnostic signals can resist causal control. Weaker, checkable causal claims are more defensible.
[C4] Formula leakage as a core shortcut type. “Formula leakage” was one of five primary shortcut families.	Demoted to an appendix sanity check only. Not a core contribution.	Formula leakage reduces to lexical retrieval, not scientifically interesting shortcut reasoning. Including it invites reviewer dismissal.

Table 1 continued.

Original design	Revised design (PhysMon)	Reason for change
[C5] 250–400 triplets, five shortcut types, 2–3 domains. Broad scope with high artefact risk.	120–180 canonical families, two core cue types (irrelevant variable; unit-compatible distractor), two domains (mechanics; electrostatics / circuits), with optional frame-rendering cue. Thermodynamics excluded from v1.	Depth over breadth for a secondary project. Thermodynamics verbal assumptions make clean invariance certificates harder.
[C6] Randomised-label control as main artefact check. Near-chance on shuffled labels treated as ruling out artefacts.	Mandatory suite of 20+ controls: TF-IDF, cue-only, prompt-statistics, length-matched, formula-count-matched, variable-count-matched, paraphrase-matched, same-answer/no-cue, variable-renaming controls, and family-level held-out splits.	Near-chance on shuffled labels only rules out trivial memorisation; it does not eliminate format, length, or style confounds. This is the primary remaining reviewer attack vector.
[C7] “Extract everything at every layer and token.” Framing implied exhaustive extraction.	Staged, hypothesis-driven hierarchy: cue-token residual stream, last-prompt-token residual stream, attention output, MLP output. The strongest intended result is prompt-side activations predicting sensitivity before generation begins.	Exhaustive extraction has no scientific priority ordering and produces excessive storage costs; reviewer-attack surface increases with number of comparisons.
[C8] Generated-reasoning-token patching as a core method. Patching at “reasoning-token positions” was central.	Generated-reasoning-token patching demoted to an optional extension requiring teacher-forced or aligned positions. Core causal work uses prompt-side sites only.	Independently sampled reasoning traces are semantically misaligned; patching across them is methodologically unsound.

Table 1 continued.

Original design	Revised design (PhysMon)	Reason for change
[C9] False-negative rate as primary and only headline metric.	AUROC and AUPRC on held-out template families are primary. FNR is a derived result at a pre-registered fixed FPR only. Calibration (Brier score, ECE) added.	FNR alone is threshold-sensitive and depends on how the sensitivity label is binarised. AUROC is standard and threshold-independent.
[C10] Month-by-month timeline. Six-month timeline with calendar months.	11-stage ordered execution plan with explicit decision gates and no dates, durations, or deadlines.	Calendar timelines conflate scientific ordering with time availability. Decision gates are scientifically more meaningful.
[C11] Reliance on commercial API credits. GPT-4o and Claude 3.5 used for generation and verification; no alternative specified.	All analysis uses locally hosted open-weight models on the Sharanga cluster (H200, H100, A100 GPUs). No commercial API credits required. A Qwen3.5-397B-class model is used for behavioural and judge roles only, not activation patching.	Commercial API reliance introduces cost uncertainty and reproducibility concerns. The open-weight strategy is a reproducibility strength.
[C12] Weak baselines. Three baselines: CoT classifier, logprob confidence, self-consistency.	Comprehensive baseline suite: surface, behavioural, textual-monitor, and causal-control tiers, including generic correctness probe and black-box two-query counterfactual baseline.	2025–2026 venue standards require strong baselines including process critics, judge ensembles, and generic correctness comparators.
[C13] LLM-generated physics and ground truth. LLMs used to generate problem physics.	Symbolic-template-first pipeline. LLMs may paraphrase and search for adversarial artefacts only. All physics and ground-truth labels come from templates, solvers, and human validation.	LLM-generated physics problems frequently contain subtle assumption drift; invariance certificates require formal verification.

Table 1 continued.

Original design	Revised design (PhysMon)	Reason for change
[C14] No formal problem definition or notation.	Formal notation section defining $\mathcal{F}_\tau$, $S_\theta(\tau)$, $\hat{S}_\theta(\tau)$, $S_\theta^{\text{lp}}(\tau)$, and the three-level framework.	Precise definitions are required for a main-track submission and prevent definitional slippage throughout the paper.
[C15] No go/no-go criteria.	Seven explicit go/no-go criteria with three-tier thresholds (proceed / redesign / stop-pivot) for each of: behavioural sensitivity, dataset invariance, artefact control, hidden-state advantage, intervention effect, replication, and correctness-probe residualisation.	Without go/no-go criteria the project risks continuing past signal failure and overinterpreting weak results.

2 Unresolved Decisions Requiring Investigator Choice

The following decisions were deliberately left open because they require domain expertise, resource assessment, or empirical pilot results that are not available at proposal-writing time.

1. **[D1] Model selection: specific versions.** The proposal specifies model *families* (one Qwen-family dense 7–14B; one Llama- or Mistral-family 7–14B; one reasoning-tuned open model; one large model for behavioural/judge roles). The *specific* model versions should be selected immediately before Stage 2 instrumentation based on: current availability on the Sharanga cluster, confirmed TransformerLens or equivalent hook support, licensing, and reasoning strength on a physics hold-out. **This is Stage 2 decision; do not defer past that point.**
2. **[D2] Sensitivity measure for binary labelling.** The proposal defines both $\hat{S}_\theta(\tau)$ (answer-flip rate) and $S_\theta^{\text{lp}}(\tau)$ (maximum log-probability drop) for binary or continuous targets. A decision is needed on: which measure is the primary probing target; what threshold converts continuous sensitivity to a binary go/no-go family label; and whether the threshold is pre-registered before any hidden-state analysis or estimated from the training families. **Pre-register the threshold before Stage 5.**
3. **[D3] Paraphrase generation strategy.** The proposal requires at least two controlled paraphrases per family. A decision is needed on: which locally hosted model generates paraphrases (a local Qwen- or Llama-family model is a natural choice), how many

paraphrase passes are reviewed by a human validator, and whether variable renaming is applied algorithmically or manually. **Decide and document before pilot generation begins in Stage 3.**

4. **[D4] Answer-parser design.** The proposal requires a normalised answer parser that extracts the canonical answer from free-text model output. Options include: rule-based extraction for numeric answers, a lightweight classifier for symbolic expressions, or a parser model. The parser must be validated on held-out examples before any sensitivity measurement. **Specify and validate in Stage 2.**
5. **[D5] Number of numeric instantiations per family.** The proposal states m variants per canonical family. The exact value of m (at least 2, practically 3–5) and whether m varies by cue type must be specified. Larger m increases sensitivity estimation precision but multiplies annotation effort. **Fix m before the pilot in Stage 3 and document the choice.**
6. **[D6] Teacher-forcing protocol for generated-token causal work (Stage 8 extension).** If the investigators decide to include generated-token patching as a Stage 8 extension, a specific teacher-forcing alignment scheme must be documented before any intervention experiment starts. The proposal notes this is optional and requires demonstrated alignment.
7. **[D7] Venue submission order.** The proposal presents ACL/EMNLP/NAACL as primary targets and ICLR as a stretch target. The specific submission order (e.g., submit to EMNLP 2026 first; pivot to ACL 2027 if rejected; consider ICLR 2027 if causal evidence is strong) must be decided after Stage 10 (claim freeze), using the actual evidence in hand.
8. **[D8] Benchmark release policy.** The proposal prohibits public benchmark release before the core analysis paper is internally complete. The *public release date* relative to paper submission (simultaneous with preprint, after acceptance, after proceedings publication) is a community-norms and strategic decision that the investigators must make before Stage 11.
9. **[D9] Human validator recruitment.** The proposal requires dual validation by people with appropriate physics competence (undergraduate degree or higher in a physics-related field, or confirmed advanced coursework) for all development and test families. Identifying, recruiting, and training these validators, and specifying their compensation and adjudication process, is an operational decision.
10. **[D10] Reasoning-tuned model intervention scope.** The proposal designates one reasoning-tuned open model for external validity only. If at Stage 7 it becomes clear that the hidden-state advantage is strong and the reasoning-tuned model has stable activation tooling, the investigators may want to extend the causal Stage 8 work to that model. This escalation requires an explicit decision and additional compute estimation before Stage 8.

3 Assumptions Made During Revision

The following assumptions were made during proposal revision. Each should be verified by the investigators before work begins.

1. **[A1] Sharanga cluster access is confirmed.** The proposal assumes access to the Sharanga cluster with the specified GPU allocation ($2\times$ H200, $3\times$ H100, $2\times$ A100). If the allocation changes materially, the model selection and compute budget sections must be revised.
2. **[A2] TransformerLens or equivalent hooking support exists for the selected models.** The proposal assumes stable activation extraction and patching is achievable using existing open-source tooling. If a selected model has non-standard architecture that breaks hook support, a different model must be chosen at Stage 2.
3. **[A3] No near-identical published paper exists at submission time.** The proposal treats “PhysMon” as a real gap in the literature as of mid-2026. The literature map (Section 2 of the proposal) was based on the source review, which itself has a knowledge cutoff. A fresh literature search *immediately before* benchmark development begins is strongly recommended.
4. **[A4] Classical mechanics and electrostatics/circuits admit clean symbolic invariance certificates.** The proposal assumes that symbolic solvers (e.g., SymPy) can certify answer invariance for the planned cue families. If a cue type resists clean symbolic specification, it must be redesigned or dropped before Stage 6.
5. **[A5] Log-probability access is stable across models.** The JSD-based sensitivity measure $S_\theta(\tau)$ requires access to log-probabilities over the answer vocabulary. Some inference configurations suppress this. Verify in Stage 2 before committing to log-probability-based metrics.
6. **[A6] Dual validators with appropriate physics background are available.** The proposal assumes two qualified human validators can be identified for every development and test family. If this proves infeasible, an alternative (e.g., tiered validation with a full review for test families only) must be designed and documented.
7. **[A7] The bibliography arXiv IDs are correct for all entries except the flagged ones.** Two bib entries have flagged discrepancies: `zhang2025correctness` (the arXiv ID 2305.15054 appears misattributed in the source review) and `sahoo2022probes` (the 2022 arXiv prefix conflicts with the 2026 year in the review). All other arXiv IDs were taken directly from the review document and should be individually verified against the actual arXiv pages before submission.
8. **[A8] The Qwen3.5-397B-class model is available for local inference.** The proposal uses a very large open-weight model for behavioural and judge roles. If this model is unavailable locally, a smaller but strong locally hosted model or a restricted number of commercial API calls (if later available) can substitute for the behavioural/judge role only—not for the core activation analysis.
9. **[A9] The sensitivity signal exists at pilot scale.** The entire project depends on H1: that non-trivial cue sensitivity $\hat{S}_\theta(\tau) > 0$ appears in at least 20% of pilot families in at least one primary model. This is an empirical assumption that the pilot study will either confirm or refute. If it fails, the go/no-go criteria mandate stopping or major redesign.

4 Compilation Notes and Required Packages

4.1 Environment

The proposal was compiled with `pdflatex` under TeX Live 2023 (Debian package).

4.2 Required Packages

All packages are available in a standard TeX Live installation.

Package	Purpose
<code>geometry</code>	Page margins
<code>inputenc (utf8)</code>	Input encoding
<code>amsmath, amssymb, mathtools</code>	Mathematical notation
<code>booktabs</code>	Professional table rules
<code>tabularx</code>	Fixed-width flexible-column tables
<code>longtable</code>	Multi-page tables (used for lit map, go/no-go, and baselines)
<code>array, multirow</code>	Column types and row spanning
<code>enumitem</code>	Compact list formatting
<code>xcolor</code>	Colour definitions
<code>hyperref</code>	Hyperlinked cross-references and citations
<code>natbib</code>	Author–year citation format (<code>round</code> , <code>sort&compress</code>)
<code>setspace</code>	Line-spacing control

4.3 Packages Intentionally Omitted

- `lmodern`: not installed in the compilation environment. Removed from the source; the document compiles with Computer Modern type 1 fonts (`cm*.pfb`). Restore by adding `\usepackage{lmodern}` if your TeX distribution includes it.
- `microtype`: removed because font expansion requires scalable fonts (T1 encoding + LM or equivalent). Can be restored together with `\usepackage[T1]{fontenc}` and `\usepackage{lmodern}` on a full installation.
- `fontenc (T1)`: removed together with `microtype` for the same reason.

4.4 Compilation Sequence

```
pdflatex physmon_proposal.tex
bibtex   physmon_proposal
pdflatex physmon_proposal.tex
pdflatex physmon_proposal.tex
```

Two `pdflatex` passes after `bibtex` are required to resolve all cross-references, the table of contents, and the `longtable` continuation headers.

4.5 Output

The compiled PDF is 30 pages and approximately 266 KB. A table of contents is generated automatically. All internal references (`\ref`, `\label`, `\citet`, `\citep`) resolve without undefined-reference warnings after the full compile sequence.

4.6 Bibliography Discrepancy Flags

Two bibliography entries contain **WARNING** notes in the `note` field. These will appear in the compiled reference list. They should be resolved before any public or submission version:

- **zhang2025correctness**: arXiv:2305.15054 (attributed in the source review) is the Lightman et al. 2023 process reward model paper “Let’s Verify Step by Step,” not “Reasoning Models Know When They’re Right” by Zhang et al. 2025. The correct arXiv ID must be identified and the author list verified.
- **sahoo2022probes**: the arXiv prefix 2207.* indicates a July 2022 submission, but the review describes the paper as a 2026 contribution titled “Linear Probes Detect Task Format, Not Reasoning Mode” by Sahoo et al. Verify whether this is an extended version published in 2026, or whether the arXiv ID is incorrect.

Several other entries use placeholder author lists (“et al.”) because full author rosters were not available during revision. These must be replaced with complete author names before submission.

5 Self-Audit: Review Criticisms Addressed

This section maps each major criticism from the deep-research review to the exact revised section that addresses it. Severity labels match the review’s own classification (Fatal / Manageable).

Table 2: Review criticism → revised-proposal response.

Review criticism	Severity	How and where addressed in PhysMon
Prompt condition is not model reasoning path. Probe predicts cue presence, not shortcut-reliant computation.	Fatal unless redesigned	§3 (Formal Problem Definition) introduces the three-level framework distinguishing prompt condition, observed behaviour, and internal representation. §4 (Hypotheses) makes the probe target explicitly model-specific counterfactual sensitivity, not cue presence. §9 (Probes) specifies behavioural labels only. §13 (Go/No-Go criterion 7) requires residualisation over difficulty and correctness.
Prompt artefacts can explain probe performance. Format, length, style, cue tokens drive probe separation.	Fatal unless controlled	§7 (Artefact-Control Protocol) mandates 20+ controls including TF-IDF, cue-only, prompt-statistics, length-matched, formula-count, variable-count, unit-matched, style-matched, same-answer/no-cue, and variable-renaming controls. §10 (Baselines) lists all surface baselines as mandatory. §12 (Go/No-Go criterion 3) sets a hard artefact-control threshold.
Randomised-label control is too weak. Near-chance on shuffled labels proves almost nothing.	Manageable	§7 replaces the single shuffle control with a comprehensive matched and held-out protocol. §5 (Generation Protocol) includes adversarial artefact search. Randomised label is retained only as a weak sanity check, not the primary control.
Formula leakage is trivial. If formula is present, the task reduces to lexical retrieval.	Manageable if demoted	Formula leakage completely demoted from a core shortcut family to an appendix sanity check (§20, Appendix A). The main text does not include formula leakage as a shortcut type.

Table 2 continued.

Review criticism	Severity	How and where addressed in PhysMon
Current labels denote prompt condition, not shortcut use. “Shortcut-available” label encodes cue insertion, not reliance.	Fatal unless changed	§3 precisely defines that the probe target is $S_\theta(\tau)$ or $\hat{S}_\theta(\tau)$, which are behavioural measures of model output change across an invariant family. The term “shortcut reliance” is explicitly reserved for cases supported by intervention evidence (§3.4, Terminology).
“Same answer but different reasoning path” is not observable. Internal paths are not directly observable.	Manageable	All path language removed from the proposal. The framework discusses equivalent-task families, output invariance, output instability, and internal representations—never unobservable reasoning paths. This is explicit in §3.1 (Three Levels of Analysis).
Activation patching does not establish the claimed causal result. Patching changes outputs without transferring shortcut reasoning.	Fatal unless narrowed	Causal claim narrowed to: “localised representations contribute to cue sensitivity” (§11.1). Necessity and sufficiency tests are required (§11.2–11.3). Mandatory controls include norm-matched random direction, random site, nearby layer, unrelated family, and same-answer/no-cue donor (§11.4). H4 is explicitly labelled higher-risk with a documented fallback (§4.4, §11.5).
Generated-token patching is poorly aligned. Reasoning-token positions are semantically misaligned across runs.	Manageable	Generated-reasoning-token patching demoted to an optional extension in §8.1. Core causal work uses prompt-side positions only. Teacher-forcing requirement noted if generated-token work is attempted.

Table 2 continued.

Review criticism	Severity	How and where addressed in PhysMon
Baselines are underpowered for 2026 standards. No process critics, judge ensembles, or generic correctness probes.	Fatal for main-track	§10 provides a comprehensive baseline table with 20+ required baselines in four tiers: surface, behavioural, textual-monitor, and causal-control. Generic correctness probe is mandatory. Two-query black-box counterfactual is mandatory. Judge ensemble (two settings or models) is mandatory. PRM/critic baseline is strongly recommended.
Primary metric is too narrow and threshold-sensitive. FNR depends on binarisation threshold.	Manageable	AUROC and AUPRC on held-out template families are now primary metrics (§12.1). FNR is reported only at a pre-registered fixed FPR as a derived result. Brier score and calibration error added.
Dataset scope too broad for a secondary project. 250–400 triplets, five shortcut types, three domains, three models, full patching is too much.	Manageable (strongly recommended)	Scope reduced to 120–180 canonical families, two cue families, two domains (mechanics; electrostatics/circuits), two primary dense models plus one reasoning-tuned replication model. Thermodynamics excluded from v1. Formula leakage removed from core. All deferred scope items listed in §18 (Extensions).
7B-only model set may weaken impact. Strong reasoning models are relevant benchmarks.	Medium–High	Three-tier model strategy: two primary dense 7–14B models for full internal analysis; one reasoning-tuned open model (7–32B) for external validity; one large model (~397B class) for behavioural and judge roles (§6.2).

Table 2 continued.

Review criticism	Severity	How and where addressed in PhysMon
Framing split across too many identities. Simultaneously mechanistic, CoT faithfulness, shortcut, and physics benchmarking without committing.	Manageable	Primary framing consolidated as: <i>counterfactual scientific-reasoning monitorability with limited causal representation analysis</i> . Venue strategy section (§16) explicitly separates ACL/EMNLP/NAACL framing (monitoring) from ICLR stretch framing (causal).
LLM-assisted generation plus single-layer verification is risky. LLM-generated physics may contain assumption drift.	Potentially fatal	Symbolic-template-first pipeline mandated (§5.2). LLMs restricted to paraphrasing, variable diversification, and adversarial artefact search. All physics and labels come from templates, solvers, and human validation. Dual human validation required for all development and test families (§7.1).
Judge-based faithfulness evidence is inherently noisy. Faithfulness scores change materially with judge/classifier choice.	Manageable	At least two judge models or prompt settings required (§10). Judge sensitivity (performance range across choices) reported explicitly. No single judge treated as ground truth. §10 notes this explicitly.
No statistical analysis plan. No mention of multiple-testing correction, effect sizes, or appropriate inference units.	Not in original severity table; implicit	§12.4 (Statistical Analysis) specifies: bootstrap CIs over canonical families, paired randomisation tests, BH correction for layer-wise scans, effect sizes with CIs, calibration uncertainty, seed sensitivity, and pre-specified primary layers. Canonical family is the inference unit throughout.
No formal notation or problem definition. Constructs were informal; labels potentially circular.	Not in original severity table; implicit	§3 provides complete formal notation: template τ , family $\mathcal{F}_\tau$, sensitivity measures S_θ , $\hat{S}_\theta$, S_θ^{lp} , solver-verified invariance condition, and the three-level analysis framework.

Table 2 continued.

Review criticism	Severity	How and where addressed in PhysMon
No go/no-go criteria. Project could continue past signal failure without explicit stopping rules.	Not in original severity table; implicit	§13 contains a seven-criterion go/no-go table with three-tier thresholds (proceed / redesign / stop-pivot) for: behavioural sensitivity, dataset invariance, artefact control, hidden-state advantage, intervention effect, cross-model replication, and difficulty residualisation.